

Yoshua Bengio: Deep Learning Cognition

Full Keynote Summary — AI in 2020 & Beyond (RE•WORK Keynote)

KEYNOTE HIGHLIGHT

Executive Overview: Turing Award Laureate Prof. Yoshua Bengio outlines a vision for the future of AI. While deep learning has achieved remarkable success in perception (vision, speech), current systems lack higher-level cognition, systematic generalization, and causal reasoning. Bengio proposes bridging current **System 1 (unconscious, intuitive)** deep learning with **System 2 (conscious, sequential, explicit)** capabilities using principles from neuroscience and cognitive psychology.

1. Current Deep Learning Capabilities & Limitations

Deep learning has yielded revolutionary advancements in perceptual tasks, including computer vision, speech recognition/synthesis, and game-playing (e.g., Go). However, Bengio emphasizes that scaling up dataset sizes, model parameters, and compute speed alone will **not** achieve human-level intelligence. Current models exhibit fundamental weaknesses:

- **Poor Sample Complexity:** Modern deep neural networks require thousands or millions of examples to learn a new task, whereas humans learn rapidly from very few examples by leveraging prior knowledge.
- **Lack of Autonomous Feature Discovery:** Current systems rely heavily on humans to define high-level concepts through labeled data (e.g., categorizing pixels as 'cat' or 'dog'). In contrast, infants discover intuitive physics (gravity, solid objects, pressure) completely autonomously without explicit instruction.
- **Fragility & Superficial Shortcuts:** Models rely on statistical correlations rather than true understanding. For instance, a model trained on cows in green pastures fails to recognize a cow when placed on a beach because it relies on background color as a shortcut.
- **Failure under Out-of-Distribution (OOD) Data:** Standard machine learning assumes training and test data come from the same distribution. Humans excel at generalizing to counterfactual or novel scenarios (e.g., science fiction stories or rare driving hazards) by recombining familiar concepts.

2. Bridging System 1 & System 2 Cognition

To move beyond purely perceptual tasks, AI research must draw inspiration from Daniel Kahneman's cognitive framework, which divides human cognition into two distinct systems:

Feature	System 1 Cognition (Unconscious / Fast)	System 2 Cognition (Conscious / Slow)
Nature	Fast, automatic, intuitive, parallel	Slow, deliberate, sequential, logical
Verbalization	Non-verbal, implicit, hard to explain	Verbalizable, explicit, can be communicated
AI Equivalent	Current Deep Learning (perception, intuition)	Symbolic AI / Logic (GOFAI) & Reasoning
Human Example	Driving on a familiar road, recognizing a face	Navigating a new city without GPS, doing long division
Key AI Challenge	Suffer from brittleness under distribution shifts	Traditional rule-based search fails due to exponential complexity

Bengio points out that traditional AI (GOFAI) tried to implement System 2 logic directly using hardcoded rules and massive tree searches, which failed because it lacked System 1 intuition. Human experts (e.g., chess grandmasters) do not evaluate millions of moves; they combine System 1 intuition (evaluating 2-3 promising moves instantly) with System 2 conscious planning.

3. Core Pillars for High-Level Cognitive AI

A. The Consciousness Prior & Soft Attention

Consciousness acts as a low-dimensional bottleneck that selects a tiny subset of concepts from an immense high-dimensional unconscious state. **Attention** mechanisms allow models to focus dynamically on specific elements sequentially. Unlike hard attention, soft attention uses continuous, differentiable competition scores, allowing neural networks to learn where to attend end-to-end via backpropagation.

B. Episodic Memory & Credit Assignment Through Time

Standard Backpropagation Through Time (BPTT) requires unrolling neural states over long sequences, which is computationally expensive and biologically implausible. Bengio proposes leveraging episodic memory (inspired by the hippocampus) to jump backwards in time to key causal events—functioning like a 'time machine' to perform targeted credit assignment (e.g., remembering hearing a tire pop hours before discovering a flat tire).

C. Disentangled Representations & Affordances

High-level concepts should be represented as disentangled, independent factors. Psychologically, entities are defined by **affordances**—how an agent interacts with them (e.g., a chair is something you sit on; a door is something you open) rather than low-level physical traits like color or shape. Unsupervised agents should learn representations organized around aspects of the environment they can actively control.

D. Causality & Interventions

Machine learning usually models joint statistical dependencies. However, statistical correlations break when environmental distributions shift. Causality introduces structural cause-and-effect relationships (A causes B). When an agent intervenes (e.g., turning off a light switch), a causal model adapts

instantly by modifying only the local variable affected, rather than retraining the entire network from scratch.

E. Dynamic Compositionality & Modular Architectures

Current deep networks have static computational graphs where every layer passes fixed vectors to the next. Human System 2 reasoning uses **dynamic compositionality**—stitching together reusable modules on the fly depending on the context. Bengio highlights architectures like **Recurrent Independent Mechanisms (RIMs)** and Transformers, which process sets of objects dynamically using attention, drastically improving out-of-distribution generalization.

F. Grounded Language Learning

Language models trained exclusively on text lack world models and common-sense grounding, making them prone to naive semantic errors. True language understanding requires grounded language learning, where agents learn text alongside perception and action within interactive environments.

4. Summary & Future Outlook

FINAL TAKEAWAY

Conclusion: Deep learning is not dead, nor is it complete. The path toward human-level AI requires expanding the deep learning toolbox by combining System 1 perception with System 2 high-level cognition. By incorporating **attention, episodic memory, causal world models, and dynamic modularity**, future AI systems will generalize beyond their training distributions and autonomously build common-sense understanding.